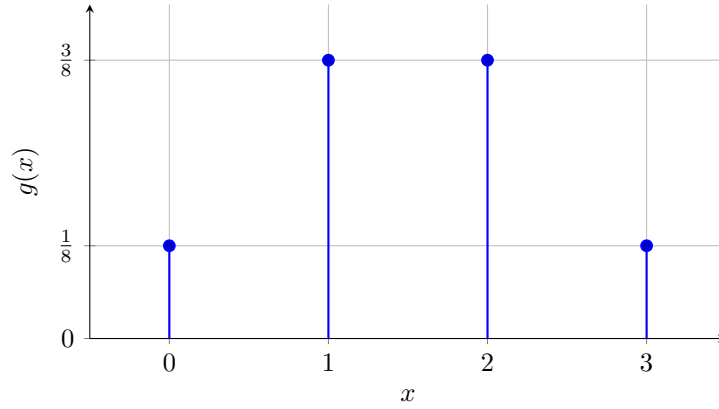


We have seen in Example 2.2 that Y is a discrete RV with support $S = \{0, 1, 2, 3\}$. The PMF of Y is $g : \mathbb{R} \rightarrow \mathbb{R}$ where

$$\begin{aligned} g(0) &= G(0) - G(0-) = \frac{1}{8} - 0 = \frac{1}{8} \\ g(1) &= G(1) - G(1-) = \frac{1}{2} - \frac{1}{8} = \frac{3}{8} \\ g(2) &= G(2) - G(2-) = \frac{7}{8} - \frac{1}{2} = \frac{3}{8} \\ g(3) &= G(3) - G(3-) = 1 - \frac{7}{8} = \frac{1}{8}. \end{aligned}$$

Thus the PMF of Y is DF

$$g(x) = \begin{cases} \frac{1}{8}, & \text{if } x = 0, 3 \\ \frac{3}{8}, & \text{if } x = 1, 2 \\ 0, & \text{Otherwise.} \end{cases}$$



Example 2.4. A fair die (all outcomes are equally likely) is tossed repeatedly and independently until a 6 is observed. Then X is a discrete RV with support $S = \{1, 2, \dots, 6\}$. The PMF

$$f(x) = \mathbb{P}(X = x) = \begin{cases} \left(\frac{5}{6}\right)^{x-1} \frac{1}{6}, & \text{if } x = 1, 2, \dots, 6 \\ 0, & \text{Otherwise.} \end{cases}$$

and DF

$$F(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & \text{if } x < 1 \\ \frac{1}{6}, & \text{if } 1 \leq x < 2 \\ \frac{11}{36}, & \text{if } 2 \leq x < 3 \\ \sum_{j=1}^i \left(\frac{5}{6}\right)^{j-1} \frac{1}{6} = 1 - \left(\frac{5}{6}\right)^i, & \text{if } i \leq x < i+1 \\ \vdots & \end{cases}$$

3 Continuous random variable

Let X be a RV with DF F .

Definition 3.1. The RV X is said to be a continuous RV if there exists a non-negative integral function $f : \mathbb{R} \rightarrow [0, \infty)$ such that, for any $x \in \mathbb{R}$,

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt.$$

The function $f(\cdot)$ is called the probability density function (PDF) of X . The support of the continuous RV X is the set $S = \{x \in \mathbb{R} : F(x+h) - F(x-h) > 0, \forall h > 0\}$.

Remark 3.1. (1) From the fundamental theorem of calculus we know that the definite integral

$$F(x) = \int_{-\infty}^x f(t)dt$$

is a continuous function of $\mathbb{R}$. Thus the DF of any continuous RV X is continuous everywhere on $\mathbb{R}$. In particular

$$\mathbb{P}(X = x) = F(x) - F(x-) = 0, \forall x \in \mathbb{R}.$$

Generally if A is any countable subset of $\mathbb{R}$ then for any continuous RV X

$$\mathbb{P}(X \in A) = \sum_{x \in A} \mathbb{P}(X = x) = 0.$$

(ii) There are random variables that are neither discrete nor continuous (see Example 1.4). Some random variables will not be stated here.

(iii) If X is a continuous RV then

$$\begin{aligned} \mathbb{P}(X < x) &= \mathbb{P}(X \leq x) = F(x), \forall x \in \mathbb{R} \\ \mathbb{P}(X \geq 2) &= 1 - \mathbb{P}(X < x) = 1 - F(x), \forall x \in \mathbb{R}, \end{aligned}$$

and for $-\infty < a < b < \infty$,

$$\begin{aligned} \mathbb{P}(a < X < b) &= \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X \leq b) \\ &= F(b) - F(a) \\ &= \int_{-\infty}^b f(t)dt - \int_{-\infty}^a f(t)dt = \int_a^b f(t)dt. \end{aligned}$$

(iv) Let $f(\cdot)$ be the PDF of a continuous RV X and let $E \subseteq \mathbb{R}$ be any countable subset of $\mathbb{R}$. Define $g : \mathbb{R} \rightarrow [0, \infty)$ by

$$g(x) = \begin{cases} f(x), & \text{if } x \in \mathbb{R} \cap E^c \\ C_x, & \text{if } x \in E \end{cases}$$

where $C_x \geq 0$ are arbitrary. Then

$$F(x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^x g(t)dt, \forall x \in \mathbb{R}$$

and thus g is also a PDF of X . Thus the PDF of a continuous RV is not unique.

We state the following theorem without proof.

Theorem 3.1. Let X be a RV with DF F . Suppose that F is differentiable everyone except on a countable set E . Further suppose that

$$\int_{-\infty}^{\infty} F'(t)dt = 1.$$

Then X is a continuous RV with PDF

$$f(x) = \begin{cases} F'(x), & \text{if } x \in E^c \\ 0, & \text{if } x \in E \end{cases}$$

Remark 3.2. (a) The PDF determines the DF uniquely. Converse is not true. However, the DF determines the PDF almost uniquely (they may vary in sets that have no length or have zero content). Thus it is enough to study the PDF of a continuous RV.

(b) Let X be a continuous RV with PDF $f(\cdot)$. Then

- (i) $f(x) \geq 0, \forall x \in \mathbb{R}$ and
- (ii) $\int_{-\infty}^{\infty} f(t)dt = 1$.

Conversely suppose that $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that

- (i) $g(x) \geq 0, \forall x \in \mathbb{R}$ and
- (ii) $\int_{-\infty}^{\infty} g(t)dt = 1$.

Then $g(\cdot)$ is the PDF of some continuous RV having support $T = \{x \in \mathbb{R} : \int_{x-h}^{x+h} g(t)dt > 0, \forall h > 0\}$.

Example 3.1. Let X be a RV with DF

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{x}{4}, & \text{if } 0 \leq x < 1 \\ \frac{x}{3}, & \text{if } 1 \leq x < 2 \\ \frac{3x}{8}, & \text{if } 2 \leq x < \frac{5}{2} \\ 1, & \text{if } x \geq \frac{5}{2}. \end{cases}$$

Examine whether X is continuous RV or a discrete RV or none?

Solution: Let D be the set of discontinuity points of F . Then $D = \{1, 2, \frac{5}{2}\}$. Since $D \neq \emptyset \implies X$ is not a continuous RV.

$$\sum_{x \in D} [F(x) - F(x-)] = \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{3}{4} - \frac{2}{3}\right) + \left(1 - \frac{15}{16}\right) = \frac{11}{48} < 1.$$

$\implies X$ is not a discrete RV. Thus is neither a discrete RV nor a continuous RV.

Example 3.2. Let X be a RV with DF

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{x^2}{2}, & \text{if } 0 \leq x < 1 \\ \frac{x}{2}, & \text{if } 1 \leq x < 2 \\ 1, & \text{if } x \geq 2. \end{cases}$$

Show that X is a continuous RV. Find the PDF of X and support of X .

Solution: Clearly F is continuous everywhere. Moreover F is differentiable everywhere except at three (countable) points 0, 1, and 2 and

$$F'(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{if } 0 < x < 1 \\ \frac{1}{2}, & \text{if } 1 < x < 2 \\ 0, & \text{if } x \geq 2. \end{cases}$$

Moreover

$$\int_{-\infty}^{\infty} F'(x)dx = \int_0^1 xdx + \int_1^2 \frac{1}{2}dx = 1.$$

$\implies X$ is a continuous RV with PDF

$$f(x) = \begin{cases} x, & \text{if } 0 < x < 1 \\ \frac{1}{2}, & \text{if } 1 < x < 2 \\ 0, & \text{Otherwise.} \end{cases}$$

The support of X is

$$\begin{aligned} S &= \{x \in \mathbb{R} : F(x+h) - F(x-h) > 0, \forall h > 0\} \\ &= \{x \in \mathbb{R} : \int_{x-h}^{x+h} f(t)dt > 0, \forall h > 0\} = [0, 2]. \end{aligned}$$

Example 3.3. Let X be a continuous RV with PDF

$$f(x) = \begin{cases} x^2, & \text{if } 0 < x < 1 \\ ce^{-x}, & \text{if } x > 1 \\ 0, & \text{Otherwise.} \end{cases}$$

where $c \geq 0$ is a constant.

(a) Find the value of c

(b) Find $\mathbb{P}(\frac{1}{2} \leq x \leq 2)$

(c) Find the support of X

(d) Find the DF of X .

Solution: (a) We have

$$\begin{aligned} \int_{-\infty}^{\infty} f(x)dx &= 1 \\ \implies \int_0^1 x^2 dx + \int_1^{\infty} ce^{-x} dx &= 1 \\ \implies \frac{1}{3} + ce^{-1} &= 1 \\ \implies c &= \frac{2e}{3}. \end{aligned}$$

(b)

$$\begin{aligned} \mathbb{P}(\frac{1}{2} \leq x \leq 2) &= \int_{\frac{1}{2}}^2 f(x)dx \\ &= \int_{\frac{1}{2}}^1 x^2 dx + \int_1^2 f(x)dx \\ &= \frac{1}{3}(1 - \frac{1}{8}) + c(e^{-1} - e^{-2}) \\ &= \frac{7}{24} + \frac{2}{3}(1 - e^{-1}). \end{aligned}$$

(c) The support of X is

$$\begin{aligned} S &= \{x \in \mathbb{R} : \int_{x-h}^{x+h} f(t)dt > 0, \forall h > 0\} \\ &= [0, \infty). \end{aligned}$$

(d) The DF of X is

$$F(x) = \int_{-\infty}^x f(t)dt.$$

For $x < 0$, clearly $F(x) = 0$. For $0 \leq x < 1$,

$$F(x) = \int_0^x t^2 dt = \frac{x^3}{3}.$$

For $x \geq 1$,

$$\begin{aligned} F(x) &= \int_0^1 t^2 dt + c \int_1^x e^{-t} dt \\ &= \frac{1}{3} + c(e^{-1} - e^{-x}) \\ &= \frac{1}{3} + \frac{2}{3}(1 - e^{-(x-1)}). \end{aligned}$$

Remark 3.3. Let X be a continuous RV with PDF $f(\cdot)$. If f is continuous at $x_0 \in \mathbb{R}$ then

$$f(x) = \lim_{\delta \downarrow 0} \int_{x-\frac{\delta}{2}}^{x+\frac{\delta}{2}} f(t) dt$$

$$\implies \mathbb{P}(x - \frac{\delta}{2} \leq X \leq x + \frac{\delta}{2}) \approx \delta f(x) \text{ for small } \delta > 0.$$

Definition 3.2. Let X be a continuous RV with DF F and support S , where S is an interval (may be unbounded). Further suppose that F is strictly increasing on S . For any $p \in (0, 1)$ the p -th quantile of the DF F is defined to be the value $\xi_p \in S$ such that $F(\xi_p) = p$ ($\mathbb{P}(X \leq \xi_p) = p$), i.e. the p -th quantile is defined by $\xi_p = F^{-1}(p)$. 0.5th-quantile is called median of F , 0.25th-quantile is called the lower quantile of F and 0.75th-quantile is called the upper quantile of F .

Example 3.4. Let X be a RV with PDF

$$f(x) = \begin{cases} \frac{x}{2}, & \text{if } 0 < x < 2 \\ 0, & \text{Otherwise.} \end{cases}$$

Find the lower quantile, the median and upper quantile.

Solution:

The given probability density function is

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

First, find the distribution function $F(x)$. For $0 < x < 2$,

$$F(x) = \int_0^x \frac{t}{2} dt = \frac{x^2}{4}.$$

Hence,

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \frac{x^2}{4}, & 0 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

The p -th quantile q_p is defined by

$$F(q_p) = p.$$

For $0 < p < 1$,

$$\frac{q_p^2}{4} = p \implies q_p = 2\sqrt{p}.$$

- **Lower quantile** ($p = \frac{1}{4}$):

$$q_{0.25} = 2\sqrt{\frac{1}{4}} = 1.$$

- **Median** ($p = \frac{1}{2}$):

$$q_{0.5} = 2\sqrt{\frac{1}{2}} = \sqrt{2}.$$

- **Upper quantile** ($p = \frac{3}{4}$):

$$q_{0.75} = 2\sqrt{\frac{3}{4}} = \sqrt{3}.$$

Thus, the lower quantile, median, and upper quantile are

$$\boxed{1, \sqrt{2}, \sqrt{3}}.$$

Module 3: Functions of a random variable and its expectation

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1 Probability distribution of a function of a discrete random variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable (RV) with distribution function (DF) F , probability mass function (pmf) f and support S . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function. Define $Z : \Omega \rightarrow \mathbb{R}$ as

$$Z(\omega) = h(X(\omega)), \omega \in \Omega.$$

Then Z is a RV and it is a function of RV X . Since we are only interested in values of RV X and Z and not the original probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We simply write $X(\omega)$, $\omega \in \Omega$ as X and $Z(\omega)$, $\omega \in \Omega$ as Z . We have

$$F(x) = \mathbb{P}(X \leq x), x \in \mathbb{R}$$

$$f(x) = \mathbb{P}(X = x), x \in \mathbb{R}$$

$\mathbb{P}(X \in S) = 1$ and $\mathbb{P}(X = x) > 0$, $x \in S$.

Define $T = h(S) = \{h(x) : x \in S\}$. Then T is countable. For any set $A \subseteq \mathbb{R}$, define $h^{-1}(A) = \{x \in S : h(x) \in A\}$. For $z \in T$,

$$\mathbb{P}(Z = z) = \mathbb{P}(h(X) = z) = \sum_{x \in S : h(x) = z} \mathbb{P}(X = x) \quad (1)$$

$$= \sum_{x \in h^{-1}\{z\}} \mathbb{P}(X = x) = \sum_{x \in h^{-1}\{z\}} f(x). \quad (2)$$

Thus $\mathbb{P}(Z = z) > 0$ for all $z \in T$ (since $\mathbb{P}(X = x) > 0$, for all $x \in S$) and $\mathbb{P}(Z \in T) = 1$ (since $\mathbb{P}(X \in S) = 1$). Further for any $z \notin T$, $\mathbb{P}(Z = z) = 0$. Thus we have the following theorem

Theorem 1.1. *Let X be a discrete random variable with support S , distribution function F and probability mass function f . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function. Then $Z = h(X)$ is a discrete RV with support $T = \{h(x) : x \in S\}$, PMF*

$$g(z) = \begin{cases} \sum_{x \in h^{-1}(\{z\})} f(x), & \text{if } z \in T \\ 0, & \text{Otherwise.} \end{cases}$$

and DF

$$G(z) = \mathbb{P}(Z \leq z) = \sum_{t \in T : t \leq z} g(t) = \sum_{\{x \in S : h(x) \leq z\}} f(x) = \sum_{x \in h^{-1}((-\infty, z]) \cap S} f(x).$$

In particular when $h : S \rightarrow \mathbb{R}$ is one-one, then

$$g(z) = \begin{cases} f(h^{-1}(z)), & \text{if } z \in T \\ 0, & \text{Otherwise.} \end{cases}$$